ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-2 **SUMMER SEMESTER, 2022-2023**

DURATION: 40 MINUTES **FULL MARKS: 15**

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. The Taylor series for a function $f(x)$ is – 6 (CO2) (PO1)

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + f'''''(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of

$$f(x) = \text{cosec}(x)$$

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(\text{cosec}(x)) = -\text{cosec}(x)\cot(x)$, and $\text{cosec}(0) =$ undefined. So we will get undefined values for $f(0)$ and $f^i(0)$.

Instead let's consider, $f(x) = \text{cosec}(x) = \dfrac{1}{\sin(x)}$, and find the Maclaurin series of the denominator $g(x) = \sin(x)$. Let's calculate its derivatives at the point $x = 0$.

$$\begin{aligned} g(x) &= \sin(x), \quad g(0) = 0 \\ g'(x) &= \cos(x), \quad g'(0) = 1 \\ g''(x) &= -\sin(x), \quad g''(0) = 0 \\ g'''(x) &= -\cos(x), \quad g'''(0) = -1 \\ g''''(x) &= \sin(x), \quad g''''(0) = 0 \end{aligned}$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$g(0+h) = g(0) + g'(0)\frac{h}{1!} + g''(0)\frac{h^2}{2!} + g'''(0)\frac{h^3}{3!} + g''''(0)\frac{h^4}{4!} + \dots$$

$$\Rightarrow g(h) = 0 + \frac{h}{1!} + 0 - \frac{h^3}{3!} + 0 + \dots$$

$$\Rightarrow g(x) = \frac{x}{1!} - \frac{x^3}{3!} + \dots \Rightarrow \sin(x) = \frac{x}{1!} - \frac{x^3}{3!} + \dots = x - \frac{x^3}{6} + \dots$$

Now, for the original function $f(x) = \text{cosec}(x)$,

$$\text{cosec}(x) = \frac{1}{x - \frac{x^3}{6} + \dots}$$

Performing algebraic long division, we get,

$$
\begin{array}{r l}
 & \frac{1}{x} + \frac{x}{6} + \dots \\
\cline{2-2}
x - \frac{x^3}{6} + \dots \Big) & 1 + 0 \\
 & 1 - \frac{x^2}{6} + \dots \\
\cline{2-2}
 & \qquad \frac{x^2}{6} + \dots \\
 & \qquad \frac{x^2}{6} + \dots \\
\cline{2-2}
 & \qquad 0 + \dots \\
 & \qquad \vdots
\end{array}
$$

Hence, we can write,

$$\Rightarrow \operatorname{cosec}(x) = \frac{1}{x} + \frac{x}{6} + \dots \qquad \text{[Quotient of the above algebraic long division.]}$$

2. Differentiate between *Interpolation* and *Extrapolation*. With suitable real-world examples, briefly write when each of the methods is applicable. 2 (CO3) (PO1)

Solution:

Any valid conceptual difference will suffice. There should be at least 1 valid real-world example that is used to explain the applicability. Some differences are —

Criteria	Interpolation	Extrapolation
Position of newly estimated points	Within the distribution of the known data points.	Outside the distribution of the known data points.
Likelihood of proper estimation	Has a greater likelihood of obtaining a valid estimate, since the pattern of the data points on either side of the newly estimated point is known.	Has a lesser likelihood of obtaining a valid estimate, since it assumes that the observed trend continues for values of x outside the range of the given data points.
Applicability	• ***Computer Graphics*** – For constructing B-splines, smooth surfaces, etc. • ***Carpentry*** – For drawing curves on wooden surfaces using different apparatus *e.g.* the Spline tool. • ***Digital Image Processing*** – For increasing the resolution of images. • ***Electronics*** – For plotting transfer functions or calibration curves of electronic components *e.g.* Thermistors.	• ***Weather Forecasting*** – For predicting the temperature, humidity levels, and weather conditions of the future. • ***Stock Market*** – For estimating the future price of company shares and stocks. • ***Cricket*** – For hypothesizing the total runs that will be scored in the innings by the batting team.

Rubric:

- 1 point for a valid conceptual difference.
- 0.5 point for each valid real-world application.

3. A thermistor, also known as thermal resistor, is a semiconductor type of resistor whose resistance is strongly dependent on temperature. To measure temperature using a thermistor, the manufacturers provide users with a Temperature (T) vs. Resistance (R) calibration curve. If we measure the resistance of the thermistor using an Ohmmeter, then we can refer to the aforementioned curve and determine the corresponding temperature value. Table-1 portrays multiple recorded observations involving a thermistor. 4 (CO2) (PO1)
Determine the value of the temperature T when the thermistor resistance is $R = 754.8\Omega$ using Newton's Divided Difference method of interpolation and a second order polynomial.

Table 1: Temperature T of a thermistor as a function of its resistance R for Question-3.

Resistance, R (Ω)	**Temperature, T** ($°C$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Solution:

The 2nd order NDD interpolant is –

$$T(R) = b_0 + b_1(R - R_0) + b_2(R - R_0)(R - R_1)$$

Given, $R = 754.8\Omega$, and the 3 nearest data points that bracket it are, $R_0 = 911.3$, $R_1 = 636$, and $R_2 = 451.1$. The corresponding T values are $T(R_0) = 30.131$, $T(R_1) = 40.12$, and $T(R_2) = 50.128$ respectively.

Now, let's calculate the coefficients.

$b_0 = T(R_0) = 30.131$

$$b_1 = \frac{T(R_1) - T(R_0)}{R_1 - R_0} = \frac{40.12 - 30.131}{636 - 911.3} = -0.0362$$

$$b_2 = \frac{\dfrac{T(R_2) - T(R_1)}{R_2 - R_1} - \dfrac{T(R_1) - T(R_0)}{R_1 - R_0}}{R_2 - R_0} = \frac{\dfrac{50.128 - 40.12}{451.1 - 636} - \dfrac{40.12 - 30.131}{636 - 911.3}}{451.1 - 911.3} = 3.877 \times 10^{-5}$$

Hence,

$$\begin{aligned} T(R) &= b_0 + b_1(R - R_0) + b_2(R - R_0)(R - R_1) \\ &= 30.131 - 0.0362(R - 911.3) + 3.877 \times 10^{-5}(R - 911.3)(R - 636); \quad 451.1 \leq R \leq 911.3 \end{aligned}$$

Now that we have the NDD interpolant, we can obtain the value of $T(754.8)$.

$$\begin{aligned} T(754.8) &= 30.131 - 0.0362(754.8 - 911.3) + 3.877 \times 10^{-5}(754.8 - 911.3)(754.8 - 636) \\ &\approx 35.075°C \end{aligned}$$

4. Prove that a polynomial of degree n or less that passes through $n+1$ data points is unique. 3

(CO4)

(PO1)

Solution:

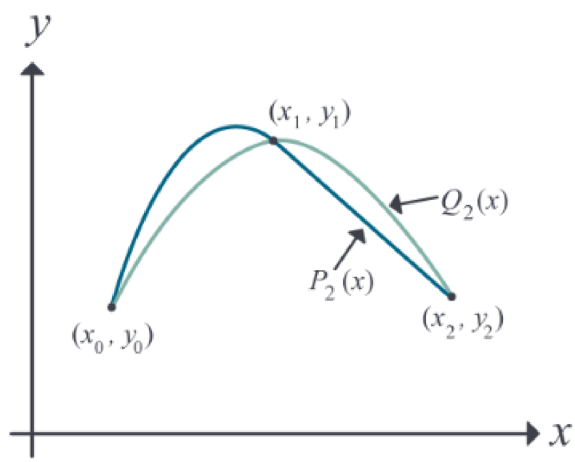

Figure 1: Interpolating polynomials through three points.

Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n+1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n+1$ data points (x_0, y_0), (x_1, y_1), $\ldots$, (x_n, y_n). Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n+1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \ldots, n\}$$

Hence

$$\begin{aligned} R_n(x_i) &= P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \ldots, n\} \\ \Rightarrow R_n(x_i) &= 0 \end{aligned}$$

The nth order polynomial $R_n(x)$ has $n+1$ zeros. A polynomial of order n can have $n+1$ zeros only if it is identical to a zero polynomial, that is,

$$\begin{aligned} R_n(x) &\equiv 0 \\ \Rightarrow P_n(x) &\equiv Q_n(x) \end{aligned}$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. $\square$ Q.E.D.

Rubric:

- 1 point for correctly defining $P_n(x)$, $Q_n(x)$, and $R_n(x)$.
- 1 point for showing $R_n(x)$ is a zero polynomial.
- 1 point for the final conclusion.